

SONAL SAI K S

Full Stack Developer

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PROFESSIONAL SUMMARY

Frontend-focused full-stack developer with **1 year 9 months** of production experience building performant, real-time web applications. Specializes in crafting responsive, accessible **React** interfaces with a strong eye for component architecture, state management, and UI performance — while remaining equally comfortable extending into backend services when the project demands it. Delivered a production VoIP transcription system integrating **Twilio Media Streams** and **Deepgram AI**, and architected event-driven microservices with **NATS** and **Python**. Thrives in fast-moving team environments, picking up new tools and codebases quickly without losing momentum. A collaborative team player who communicates clearly across design, backend, and product — turning ambiguous requirements into polished, production-ready interfaces.

TECHNICAL SKILLS

Frontend: React, TypeScript, JavaScript (ES6+), Angular, HTML5, CSS3, Redux, Context API, Vite, Framer Motion, SCSS

Backend: Node.js, Express.js, Python, FastAPI, REST API design

Real-Time: WebSockets, WebRTC, Twilio Media Streams, Deepgram AI integration, audio streaming

Databases: MongoDB, MongoDB Atlas, PostgreSQL, Redis

Infra & Tools: Docker, Git, GitHub, NATS, Postman, Vercel, Render, GitHub Pages, Chrome DevTools

PROFESSIONAL EXPERIENCE

Junior Software Engineer — AOT Technologies
Trivandrum, Kerala

June 2024 – Present

- **Real-Time Transcription Pipeline:** Architected and shipped a production VoIP transcription system handling live audio via **Twilio Media Streams**, relayed through a WebSocket pipeline to Node.js, and transcribed in real time using **Deepgram AI**
- **Frontend Development:** Developed responsive and high-performance **React** interfaces for real-time workflows, enabling live UI updates via **WebSockets** while maintaining scalable component architecture and efficient state management.
- **Real-Time Communication:** Developed a WebSocket-based bidirectional messaging layer between React clients and backend services, achieving sub-200 ms message delivery for live workflow state updates.
- **Event-Driven Architecture:** Implemented event-driven workflow execution using **Python** consumers and **NATS** messaging, enabling asynchronous task processing and scalable communication between services.

PERSONAL PROJECTS

AudioTranscription React, Vite, WebSockets, SCSS

github.com/sonalsai/AudioTranscription

- Built a real-time speech-to-text dashboard with animated recording controls, session management, and live transcript streaming via WebSockets — achieving end-to-end transcription latency under 500 ms.

SwiftDrop-Web React, WebRTC, WebSockets, Vite

github.com/sonalsai/SwiftDrop-Web

- Engineered a zero-server peer-to-peer file transfer app using **WebRTC** data channels with a custom WebSocket signaling server; supports room-based connections and browser-to-browser transfers of any file size.

EazNote React, Vite, PWA

github.com/sonalsai/EazNote

- Developed an offline-first Progressive Web App for note management with full PWA compliance (installable, service-worker caching, background sync) — fully functional with zero network dependency.

CTF Brainrot Edition React, JavaScript

github.com/sonalsai/CTF-Brainrot-Edition

- Created an interactive browser-based CTF challenge platform with debugging puzzles and UI challenges simulating real-world frontend security and problem-solving scenarios.

EDUCATION

Diploma in Computer Engineering — Maharaja's Technological Institute (MTI), Thrissur

2021 – 2024

CGPA: 7.8 / 10